

Important Formulas & Question Solving Patterns

Linear Algebra + Vector Calculus

UNIT 1: VECTOR SPACES

Important Formulas

Subspace Conditions:

1. Zero vector belongs to set
2. Closed under addition
3. Closed under scalar multiplication

Linear Combination:

$$v = a_1 v_1 + a_2 v_2 + \dots$$

Linear Dependence:

$$c_1 v_1 + c_2 v_2 + \dots = 0$$

Dimension = Number of vectors in basis

Pattern to Solve Questions

For Subspace Questions:

Check zero vector $\rightarrow$ addition $\rightarrow$ scalar multiplication.

For Linear Dependence:

Check if one vector is scalar multiple of another.

For Basis:

Check linear independence + spanning.

UNIT 2: LINEAR TRANSFORMATIONS

Important Formulas

Kernel:
 $\text{Ker}(T) = \{v : T(v)=0\}$

Range:
 $\text{Range}(T) = \{T(v)\}$

Rank-Nullity Theorem:
 $\text{Rank} + \text{Nullity} = \text{Dimension of Domain}$

Linearity Conditions:
 $T(u+v)=T(u)+T(v)$
 $T(cu)=cT(u)$

Pattern to Solve Questions

To Find Kernel:
Put $T(x)=0$ and solve equations.

To Find Nullity:
Count free variables.

To Find Rank:
 $\text{Rank} = \text{Total variables} - \text{Nullity}.$

To Check Linearity:
Verify addition and scalar multiplication.

UNIT 3: INNER PRODUCT SPACES

Important Formulas

Dot Product:

$$u \cdot v = x_1x_2 + y_1y_2 + z_1z_2$$

Norm:

$$||u|| = \sqrt{u \cdot u}$$

Angle Formula:

$$\cos \theta = (u \cdot v) / (||u|| ||v||)$$

Projection Formula:

$$\text{proj}_v u = [(u \cdot v) / (v \cdot v)]v$$

Unit Vector:

$$u / ||u||$$

Pattern to Solve Questions

For Orthogonal Vectors:

Check dot product = 0.

For Unit Vector:

Divide vector by magnitude.

For Projection:

Use projection formula directly.

For Gram-Schmidt:

Normalize first vector → remove projection from second.

UNIT 4: VECTOR & SCALAR FUNCTIONS

Important Formulas

Velocity:
 $v = dr/dt$

Acceleration:
 $a = dv/dt$

Unit Tangent Vector:
 $T = v/|v|$

Curvature:
 $\kappa = |v \times a| / |v|^3$

Gradient:
 $\nabla\phi = (\partial\phi/\partial x, \partial\phi/\partial y, \partial\phi/\partial z)$

Divergence:
 $\nabla \cdot F$

Curl:
 $\nabla \times F$

Pattern to Solve Questions

For Velocity & Acceleration:
Differentiate position vector.

For Curvature:
Find v and $a \rightarrow$ cross product $\rightarrow$ apply formula.

For Directional Derivative:
Gradient $\cdot$ unit direction vector.

For Divergence:
Differentiate components wrt same variables.

UNIT 5: VECTOR CALCULUS & THEOREMS

Important Formulas

Conservative Field:

$$\mathbf{F} = \nabla \phi$$

Irrotational Condition:

$$\nabla \times \mathbf{F} = 0$$

Green's Theorem:

$$\oint_C (Pdx + Qdy) = \iint_R (\partial Q/\partial x - \partial P/\partial y) dA$$

Gauss Divergence Theorem:

$$\oint_S \mathbf{F} \cdot \mathbf{n} dS = \iiint_V (\nabla \cdot \mathbf{F}) dV$$

Stokes' Theorem:

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n} dS$$

Pattern to Solve Questions

For Conservative Field:

Find potential function.

For Line Integral:

Parameterize curve first.

For Green's Theorem:

Find $\partial Q/\partial x$ and $\partial P/\partial y$.

For Gauss Theorem:

Find divergence then volume integral.